

# Shun Wang

PhD Student · Pre-doctoral Fellow

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## Education

### Universitat Politècnica de Catalunya (UPC)

Barcelona, Spain

PhD, Pre-doctoral Fellow (FPI)

November 2023 - Present

- Advisors: Prof. Francesc Pozo and Prof. Yolanda Vidal
- Research interests: *Structural Health Monitoring and Condition Monitoring for Wind Turbines*

### University College Dublin (UCD)

Dublin, Ireland

Visiting Researcher

February 2026 - June 2026

- Advisor: Prof. Abdollah Malekjafarian
- Research topic: *Structural Health Monitoring for Wind Turbine Blades*

### Northwestern Polytechnical University (NPU)

Xi'an, China

Master of Engineering (with distinction)

August 2016 - April 2023

- Major: Aeronautics and Astronautics Safety Engineering
- Advisor: Prof. Yongbo Li
- Scholarships and Awards: *Outstanding Master's Graduate / Thesis; Outstanding Thesis Award (CSA); National Scholarship*

### Northwestern Polytechnical University (NPU)

Xi'an, China

Bachelor of Engineering

August 2016 - July 2020

- Major: Aircraft Control and Information Engineering
- Thesis advisor: Prof. Yongbo Li
- Scholarships and Awards: *National Encouragement Scholarship; Outstanding Student Award*

## Research Experience

### Control, Data and Artificial Intelligence Group (CoDALab)

UPC, Barcelona, Spain

Advisors: Prof. Francesc Pozo and Prof. Yolanda Vidal

Nov. 2023 - Present

- Topic: *AI strategies for predictive maintenance and early damage detection for wind turbines*

### Intelligent Maintenance Group (IMLab)

NPU, Xi'an, China

Advisor: Prof. Yongbo Li

Jan. 2019 - Apr. 2023

- Topic: *Research on nonlinear characteristic analysis and fault diagnosis of rotating machinery based on entropy theory*

## Publications

\* corresponding author

([ORCID](#), [Google Scholar](#))

### Journal Articles

1. **Shun Wang\***, Yolanda Vidal, Jiacong Zhang, Francesc Pozo. Robust unsupervised gearbox fault detection under variable-speed operating conditions. *Energy and AI*, 2026, 25: 100862. [[doi](#)] [[code](#)]
2. **Shun Wang**, Yolanda Vidal\*, Francesc Pozo. Recent advances in wind turbine condition monitoring using SCADA data: A state-of-the-art review. *Reliability Engineering & System Safety*, 2026, 267: 111838. [[doi](#)]
3. **Shun Wang**, Yolanda Vidal\*, Francesc Pozo. An unsupervised approach to early fault detection and performance degradation assessment in bearings. *Advanced Engineering Informatics*, 2025, 68: 103620. [[doi](#)] [[code](#)]
4. **Shun Wang**, Yolanda Vidal\*, Francesc Pozo. Fault detection in wind turbines using health index monitoring with variational autoencoders. *Wind Energy Science*, 2025, 10: 2841–2863. [[doi](#)] [[code](#)]
5. **Shun Wang**, Yolanda Vidal\*, Francesc Pozo. Noncontact fault diagnosis of electrical equipment using modified multiscale two-dimensional color distribution entropy and thermal imaging. *International Journal of Intelligent Systems*, 2025, 2025(1): 4805844. [[doi](#)]
6. **Shun Wang**, Yongbo Li\*, Khandaker Noman, Zhixiong Li, Ke Feng, Zheng Liu, Zichen Deng. Multivariate multiscale dispersion Lempel-Ziv complexity for fault diagnosis of machinery with multiple channels. *Information Fusion*, 2024,

104: 102152. [\[doi\]](#)

7. **Shun Wang**, Yongbo Li\*, Khandaker Noman, Dong Wang, Ke Feng, Zheng Liu, Zichen Deng. Cumulative spectrum distribution entropy for rotating machinery fault diagnosis. *Mechanical Systems and Signal Processing*, 2024, 206: 110905. [\[doi\]](#)
8. **Shun Wang**, Yongbo Li\*, Jiacong Zhang, Zheng Liu, Zichen Deng. A novel feature extraction method based on symbol-scale diversity entropy and its application for fault diagnosis of rotary machines. *Structural Health Monitoring*, 2024, 23(3): 1423–1448. [\[doi\]](#)
9. **Shun Wang**, Yongbo Li\*, Shubin Si, Khandaker Noman. Enhanced hierarchical symbolic sample entropy: Efficient tool for fault diagnosis of rotating machinery. *Structural Health Monitoring*, 2023, 22(3): 1927–1940. [\[doi\]](#)
10. Yongbo Li\*, **Shun Wang**, Yang Yang, Zichen Deng. Multiscale symbolic fuzzy entropy: An entropy denoising method for weak feature extraction of rotating machinery. *Mechanical Systems and Signal Processing*, 2022, 162: 108052. [\[doi\]](#)
11. Yongbo Li\*, **Shun Wang**, Ni Li, Zichen Deng. Multiscale symbolic diversity entropy: A novel measurement approach for time-series analysis and its application in fault diagnosis of planetary gearboxes. *IEEE Transactions on Industrial Informatics*, 2022, 18(2): 1121–1131. [\[doi\]](#)
12. Yongbo Li\*, **Shun Wang**, Zichen Deng. Intelligent fault identification of rotary machinery using refined composite multi-scale Lempel-Ziv complexity. *Journal of Manufacturing Systems*, 2021, 61: 725–735. [\[doi\]](#)

## Conference Proceedings

1. **Shun Wang**, Yolanda Vidal, Francesc Pozo. Wind turbine gearbox fault detection under distribution shifts. *International Conference on Experimental Vibration Analysis for Civil Engineering Structures (EVACES 2025)*. Springer, 2025, 962–969. [\[doi\]](#)
2. **Shun Wang**, Yolanda Vidal\*, Francesc Pozo. Gearbox fault detection method under unseen working conditions and damage. *20th International Conference on Condition Monitoring and Asset Management (CM 2024)*. 2024. [\[doi\]](#)
3. **Shun Wang**, Yolanda Vidal\*, Francesc Pozo. Two-dimensional color distribution entropy: Validation and application on non-contact fault diagnosis for induction motor. *32nd European Signal Processing Conference (EUSIPCO 2024)*. 2024. [\[doi\]](#)
4. **Shun Wang**, Yongbo Li\*. Refined time-shift multiscale diversity entropy: A novel feature extraction algorithm for fault diagnosis of planetary gearbox. *14th International Conference on Damage Assessment of Structures (DAMAS 2021)*. Journal of Physics: Conference Series, 2022, 2184(1): 012010. [\[doi\]](#)
5. **Shun Wang**, Yongbo Li\*. A novel nonlinear analysis tool: Multi-scale symbolic sample entropy and its application in condition monitoring of rotary machinery. *2020 Asia-Pacific International Symposium on Advanced Reliability and Maintenance Modeling (APARM 2020)*. IEEE, 2020: 1–5. [\[doi\]](#)

## Honors & Awards

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|-----------|----------------------------------------------------------------------------------------------------|----------|
| 2025      | <b>Chinese Government Award for Outstanding Self-Financed Students Abroad</b> , CSC                |          |
| 2025      | <b>1st Place, ASCE-EMI SHM for Wind Energy Challenge</b> , ASCE-EMI & WeDoWind                     |          |
| 2024      | <b>EAWEScholarship</b> , European Academy of Wind Energy (EAWES)                                   |          |
| 2024-2027 | <b>FPI Scholarship</b> , Spanish Ministry of Science and Innovation                                | 111'758€ |
| 2024      | <b>High-Level Master's Degree Thesis Incentive Program</b> , Chinese Society of Astronautics (CSA) |          |
| 2023      | <b>Outstanding Master's Graduate / Thesis</b> , Northwestern Polytechnical University              |          |
| 2022      | <b>National Scholarship</b> , Ministry of Education of P.R.China (Top 2% in NPU)                   |          |
| 2021      | <b>National Scholarship</b> , Ministry of Education of P.R.China (Top 2% in NPU)                   |          |

## Academic Service

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### Journal Reviewer:

Information Fusion, Mechanical Systems and Signal Processing, Reliability Engineering & Systems Safety, International Journal of Mechanical Sciences, IEEE Transactions on Industrial Informatics, Nonlinear Dynamics, Structural Health Monitoring, Advanced Engineering Informatics, Expert Systems with Applications, Measurement, ISA Transactions, Pattern Recognition, IEEE Sensors Journal, Biomedical Signal Processing and Control, IEEE Access.

## Presentations

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<sup>+</sup> presenting author

1. **Shun Wang**<sup>+</sup>, Yolanda Vidal, Francesc Pozo. Unsupervised fault localization and root cause analysis for wind turbines via spatio-temporal representation learning. Oral presentation: 22nd EAWES PhD Seminar, August 31–September 4,

2026, Espoo, Finland.

2. **Shun Wang<sup>+</sup>**, Yolanda Vidal, Francesc Pozo. Unsupervised fault localization and root cause analysis for wind turbines via spatio-temporal representation learning. Oral presentation: 8th European Conference on Structural Control (ECSC 2026), August 25–27, 2026, Lyon, France.
3. **Shun Wang<sup>+</sup>**, Yolanda Vidal, Francesc Pozo. Uncertainty-aware bearing remaining useful life prediction based on conformal prediction. Oral presentation: 9th European Conference of the PHM Society (PHME 2026), July 8–10, 2026, Oslo, Norway.
4. **Shun Wang<sup>+</sup>**, Yolanda Vidal, Francesc Pozo. Wind turbine fault detection through a virtual sensor using exogenous variables. Poster: The Science of Making Torque from Wind (TORQUE 2026), June 3–5, 2026, Bruges, Belgium.
5. **Shun Wang<sup>+</sup>**, Yolanda Vidal, Francesc Pozo. Wind turbine fault detection through a virtual sensor using exogenous variables. Poster: WindEurope Annual Conference 2026, April 21–23, 2026, Madrid, Spain.
6. **Shun Wang<sup>+</sup>**, Yolanda Vidal, Francesc Pozo. Wind turbine condition monitoring using exogenous variables. Oral presentation: 21st EAWC PhD Seminar, October 1–3, 2025, Athens, Greece.
7. **Shun Wang<sup>+</sup>**, Yolanda Vidal, Francesc Pozo. Wind turbine gearbox fault detection under distribution shifts. Oral presentation: Eleventh International Conference on Experimental Vibration Analysis of Civil Engineering Structures (EVACES 2025), July 2–4, 2025, Porto, Portugal. [\[doi\]](#)
8. **Shun Wang<sup>+</sup>**, Yolanda Vidal, Francesc Pozo. Fault detection in wind turbines using health index monitoring with variational autoencoders. Oral presentation: 2025 Wind Energy Science Conference (WESC 2025), June 24–27, 2025, Nantes, France.
9. **Shun Wang<sup>+</sup>**, Yolanda Vidal, Francesc Pozo. Gearbox fault detection method under unseen working conditions and damage. Oral presentation: 20th International Conference on Condition Monitoring and Asset Management (CM 2024), June 17–20, 2024, Oxford, UK. [\[doi\]](#)
10. **Shun Wang<sup>+</sup>**, Yolanda Vidal, Francesc Pozo. Two-dimensional color distribution entropy: Validation and application on non-contact fault diagnosis for induction motor. Poster: 32nd European Signal Processing Conference (EUSIPCO 2024), Lyon, France. [\[doi\]](#)
11. **Shun Wang<sup>+</sup>**, Yongbo Li. A novel nonlinear analysis tool: Multi-scale symbolic sample entropy and its application in condition monitoring of rotary machinery. Oral presentation: 2020 Asia-Pacific International Symposium on Advanced Reliability and Maintenance Modeling (APARM 2020). [\[doi\]](#)
12. **Shun Wang<sup>+</sup>**, Yongbo Li. Refined time-shift multiscale diversity entropy: A novel feature extraction algorithm for fault diagnosis of planetary gearbox. Oral presentation: 14th International Conference on Damage Assessment of Structures (DAMAS 2021). [\[doi\]](#)

## Teaching Experience

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Fall 2021 Pattern Recognition and Analysis, Teaching Assistant

Fall 2020 Machine Learning, Teaching Assistant

Fall 2020 Pattern Recognition and Analysis, Teaching Assistant

## References

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### **Prof. Francesc Pozo**

[francesc.pozo@upc.edu](mailto:francesc.pozo@upc.edu)

Professor, Universitat Politècnica de Catalunya (UPC)

*Relationship: Ph.D. Supervisor*

### **Prof. Yolanda Vidal**

[yolanda.vidal@upc.edu](mailto:yolanda.vidal@upc.edu)

Professor, Universitat Politècnica de Catalunya (UPC)

*Relationship: Ph.D Supervisor*

### **Prof. Yongbo Li**

[yongbo@nwpu.edu.cn](mailto:yongbo@nwpu.edu.cn)

Professor, Northwestern Polytechnical University (NPU)

*Relationship: Master Supervisor and Undergraduate Mentor*

### **Prof. Abdollah Malekjafarian**

[abdollah.malekjafarian@ucd.ie](mailto:abdollah.malekjafarian@ucd.ie)

Associate Professor, University College Dublin (UCD)

*Relationship: Host Supervisor during research visit at UCD*